

# Finite Autonomous Execution, Capture and Renewal: Exact Joint Error and a Scoped Resource Law

MQM Paper A research addendum - development v0.1

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**Status.** New conditional derivation and constructor replay. Independent review and theorem-level novelty comparison are pending. This is an addendum to the IEEE-focused Paper A draft 0.1, not an automatic promotion of its theorem registry or physical evidence. No hardware result, quantum advantage or universal optimality is asserted.

## Abstract

We compile two quantum executions, two output captures, matched work-module renewals and finite exhaustion reporting into one prepared time-independent history Hamiltonian. An identity suffix yields a finite readout interval without further capture pulses or donor-selection switches in the ideal model. A final exhaustion qubit separates all incomplete prefixes from the completed word. This separation gives an exact normalized diamond distance,  $\sqrt{2p - p^2}$ , from the specified complete clock-and-body target, while tracing out the clock gives exact distance  $p$ , where  $p$  is the pre-exhaustion clock population. For the engineered spin-clock family, the uniform-window tail is an exact binomial sum. We derive a necessary-and-sufficient integer clock-size criterion and matching asymptotic size scaling within this fixed family, together with an explicit supply/mismatch upper bound. A thirteen-step finite example retains spectator and output correlations in outgoing donor registers and restores the active work module. The global supervising clock is not renewed: later static reversal undoes the episode. The construction is compiled per program, uses prepared finite registers, and does not establish a reusable universal autonomous processor or thermodynamic implementation.

**Index Terms:** Autonomous quantum control, history Hamiltonians, trace distance, diamond norm, finite quantum memory, resource bounds.

## 1. Development question and contribution boundary

The companion draft supplies separate execution, capture and renewal contracts but leaves the mechanism connecting their contacts as an assumption. Here we ask a narrower question: can a single prepared finite static generator carry a complete two-execution episode and offer a useful finite readout interval while all clocks, donors, outputs and correlations remain in the comparison domain?

The answer is affirmative for a compiled history construction with declared logical registers. This changes the architecture: a global supervisor implements a finite word containing computation, output

exchange, work-module exchange and an exhaustion marker. It does not construct the trigger for each earlier timed-pulse architecture, replenish the supervisor, or derive upstream resources. The program is encoded in supplied couplings rather than a reusable quantum instruction register. The new result is an explicit conditional episode and an exact error comparison for a specified target.

History propagation, perfect-transfer couplings, idle suffixes and unitary exchange are established ingredients [1]-[3]. Our exact error formula and clock-size criterion are new derivations in this development record, but priority and substantive novelty are not established. The size lower bound below applies only to the stated spin-clock padding family and readout convention. It is not a lower bound on all quantum controllers.

## 2. Complete domain, preparation and compiled word

Let  $X$  be an arbitrary computational input, possibly correlated with an arbitrary external reference  $Y$ . All finite body registers are collected in  $B$ , with  $X$  identified as one body factor. The additional bank is prepared in a specified pure state  $|b_0\rangle$ . In the example, the work module is  $W = (A, J)$ , where  $A$  is the computational qubit and  $J$  is a retained internal qubit. Donors  $D1$  and  $D2$  each have a matched pair of qubits.  $O1$  and  $O2$  are fresh output slots, and  $E$  is a fresh exhaustion qubit.

All declared logical bare body Hamiltonians are zero. This is an explicit degenerate-register benchmark, not an inference that physical gates or preparations cost no energy. The nonstationary supervising clock and its supplied gate-dependent interactions are resources. Extending to nondegenerate work Hamiltonians or a coherent-fuel architecture requires additional energy and reference contracts.

Supply  $m$  unitary primitive gates  $G_1, \dots, G_m$  on the body. The first  $m - 1$  preserve  $E = 0$ , and the final gate is  $X_E$ . Set  $P_0 = I$ ,  $P_j = G_j \dots G_1$  for  $j \leq m$  and  $P_j = P_m$  for  $j \geq m$ . Clock size  $N$  obeys  $N \geq m$ . All prefixes before  $m$  have  $E = 0$  on admitted inputs; every completed prefix has  $E = 1$ . Inputs with  $E$  initially unknown or correlated are outside this exact-flag premise unless separately admitted as supply error.

For the concrete thirteen-step example,  $H$  is the Hadamard gate,  $T = \text{diag}(1, \exp(i\pi/4))$ , and CNOT uses the first named qubit as control.

| Step  | Primitive gate                             | Purpose                                                |
|-------|--------------------------------------------|--------------------------------------------------------|
| 1-3   | $H(A), T(A), \text{CNOT}(A, J)$            | First supplied computation on the existing work module |
| 4     | $\text{SWAP}(A, O1)$                       | Capture the first computational output without cloning |
| 5-6   | $\text{SWAP}(A, D1A), \text{SWAP}(J, D1J)$ | Renew the complete work module from donor D1           |
| 7-9   | $T(A), H(A), \text{CNOT}(A, J)$            | Second supplied computation                            |
| 10    | $\text{SWAP}(A, O2)$                       | Capture the second output                              |
| 11-12 | $\text{SWAP}(A, D2A), \text{SWAP}(J, D2J)$ | Renew the work module from donor D2                    |
| 13    | $X(E)$                                     | Mark exhaustion of the compiled two-execution schedule |
| 14-N  | Identity                                   | Finite idle suffix; no third donor slot is available   |

Prepare  $J, D1J, D2A, D2J, O1, O2, E$  in  $|0\rangle$ ,  $D1A$  in  $|+\rangle$ , and  $A$  in the arbitrary input. The complete

initial state is  $\rho_{AY}$  tensor the prepared bank. After the ideal complete word,  $W$  is  $|00\rangle$ . The donors retain old internal-state correlations, and the outputs retain all correlations created by the specified computations. The second output need not be pure: its correlations with internal  $J$  are exported into  $D2J$ . No register is discarded to claim renewal.

This is renewal of the work module only. The supervising clock  $C$  is outside  $W$  and is not refreshed. Calling this a full clock reset or an indefinitely reusable machine would exceed the construction.

$E$  marks completion of the compiled schedule. It does not independently validate a logical answer, diagnose all physical errors or certify irreversible fuel depletion.

### 3. Static generator and one-launch execution

For rate  $\kappa > 0$ , define the engineered clock

$$H_N = \kappa \sum_{j=0}^{N-1} \sqrt{(j+1)(N-j)} (|j+1\rangle\langle j| + |j\rangle\langle j+1|).$$

With prefix dressing

$$\mathcal{V} = \sum_{j=0}^N |j\rangle\langle j| \otimes P_j, \quad H_{\text{hist}} = \mathcal{V}(H_N \otimes I)\mathcal{V}^\dagger,$$

the address edge  $j$  to  $j+1$  carries  $G_{j+1}$  for  $j < m$  and identity afterwards. All interactions are constant after the initial launch from clock address 0. Gate words and exchange routing are compiled into this one generator; no run-time classical gate pulse, capture switch or donor address selection is used in the ideal episode.

**Theorem AQ1 (compiled static episode).** On a purification  $|\psi\rangle$  of any admitted body/reference input, the ideal state is

$$|\Psi_t\rangle = \sum_{j=0}^N c_j(t) |j\rangle P_j |\psi\rangle,$$

$$c_j(t) = (-i)^j \sqrt{\binom{N}{j}} \sin^j(\kappa t) \cos^{N-j}(\kappa t).$$

At  $T = \pi/(2\kappa)$  it implements the complete word exactly. At  $\pi/\kappa$  it returns projectively to the initial body and clock address.

*Proof.* Prefix dressing is unitary and preserves the initial address-0 input because  $P_0 = I$ . Identify the  $N+1$  clock addresses with symmetric spin-weight vectors. The restriction of  $\kappa \sum_a X_a$  has the stated adjacent couplings, and its factorized spin rotation yields  $c_j$ . Unitary conjugation then gives the displayed state. At the midpoint only address  $N$  is populated, hence prefix  $P_m$  applies. At the full period only address 0 survives with common phase  $(-1)^N$ , and  $P_0$  applies. The argument extends to arbitrary mixed input and spectators by purification and reduction.  $\square$

The exhaustion record is therefore finite-horizon reporting, not an absorbing halt. A later return reverses the compiled operations, including captures, renewals and  $E$ . The statement does not concern a continuously monitored trajectory; monitoring changes the channel.

## 4. Exact complete and reduced channel errors

Let  $|\chi_t\rangle = \sum_j c_j(t)|j\rangle$  be the freely evolving comparison clock and define the complete target

$$|\Phi_t\rangle = |\chi_t\rangle \otimes P_m|\psi\rangle.$$

This target retains the supervisor clock, all outputs, outgoing donors, active body, exhaustion register and Y. It asks for a completed body uncorrelated with the specified comparison clock. This is a deliberate complete-domain target; a different allowed final clock state or retained-correlation convention defines a different problem.

The pre-exhaustion population is

$$p(t) = \sum_{j=0}^{m-1} |c_j(t)|^2.$$

Let  $E_t$  be the channel from input X into clock and body with the fixed prepared bank. Let  $F_t$  be the complete-word target channel, including its comparison clock. The distance convention is  $D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1$ . Channel distance is one half the diamond norm.

**Theorem AQ2 (exact complete-domain error).** For every pure admitted input, including every spectator extension,

$$D(|\Psi_t\rangle\langle\Psi_t|, |\Phi_t\rangle\langle\Phi_t|) = \sqrt{2p(t) - p(t)^2}.$$

Consequently,

$$\frac{1}{2} \|\mathcal{E}_t - \mathcal{F}_t\|_\diamond = \sqrt{2p(t) - p(t)^2}.$$

*Proof.* Clock address orthogonality gives overlap  $\sum_j |c_j|^2 \langle \psi | P_m^\dagger P_j | \psi \rangle$ . For  $j < m$  the final exhaustion flag makes the body states orthogonal, so these terms vanish. For  $j \geq m$ ,  $P_j = P_m$  and every term is one. The overlap is exactly  $1 - p$ . Pure-state trace distance is therefore  $\sqrt{1 - (1 - p)^2}$ . An arbitrary pure input and ancilla yields the same value, and every mixed extended input is a convex mixture of pure inputs. This supplies both the upper bound and an attaining input for the diamond norm.  $\square$

**Theorem AQ3 (exact clock-traced error).** After tracing C, the distance from the complete final body target, including retained outputs, donors and Y, is exactly  $p(t)$ . The corresponding reduced channel distance is also  $p(t)$ .

*Proof.* Clock reduction gives  $\sum_j |c_j|^2 P_j \rho P_j^\dagger$ . The  $j \geq m$  part is  $(1 - p) P_m \rho P_m^\dagger$  in  $E = 1$ . The other part has trace  $p$  and support  $E = 0$ . Subtracting the target leaves two orthogonal blocks of trace norms  $p$  and  $p$ . Their half-sum is  $p$ , for every input and spectator. The same argument gives the reduced channel diamond distance.  $\square$

The difference is thus attained exactly, rather than inferred from a loose square-root upper bound. Its scope depends on a pure prepared bank, the last-step exhaustion marker, the chosen product comparison clock and the unconditional full body target. It is not a universal statement that every complete quantum computation has square-root error in every metric. It also does not make postselected success a substitute for the unconditional guarantee.

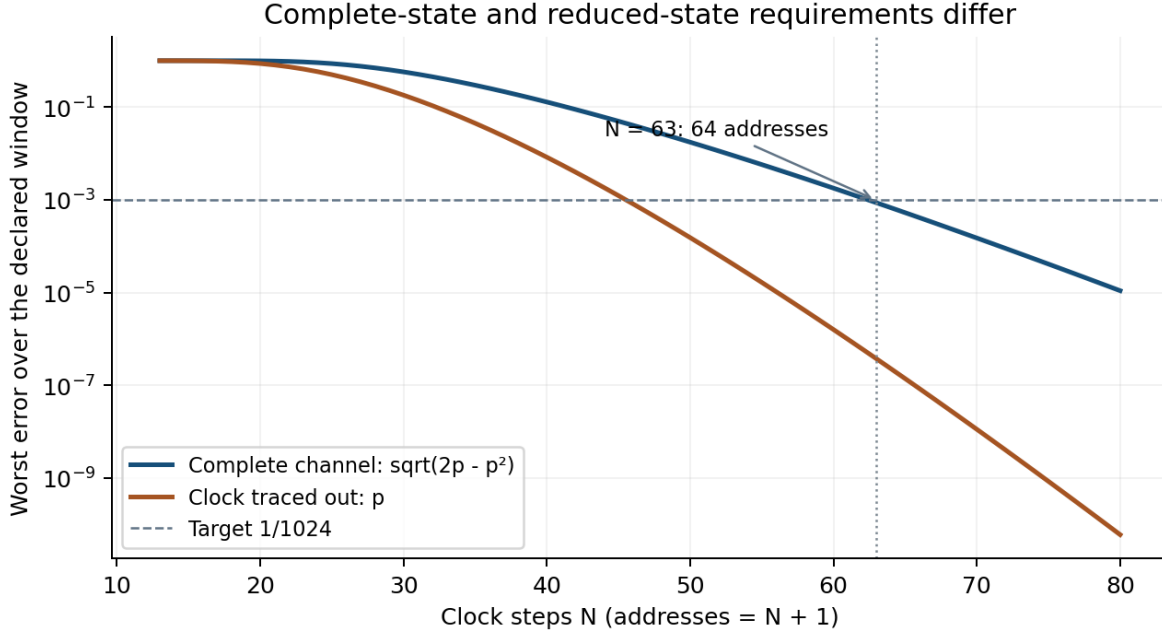


Figure 1: Exact analytic complete and reduced distances for the declared thirteen-step word. These are mathematical model values, not hardware measurements.

## 5. Finite readout window and an exact resource criterion

For  $0 < \beta < \pi/2$ , use the symmetric interval

$$I_\beta = \left[ \frac{\pi/2 - \beta}{\kappa}, \frac{\pi/2 + \beta}{\kappa} \right], \quad \mathcal{H} = \frac{2\beta}{\kappa}.$$

On this interval,  $z(t) = \sin^2(\kappa t)$  is at least  $\cos^2(\beta)$ . The clock probabilities are binomial with parameters  $N, z$ . Increasing  $z$  decreases the lower-tail probability: coupling  $N$  independent uniform random variables realizes each Bernoulli as the indicator that its uniform variable is below  $z$ , and the count is pointwise nondecreasing in  $z$ . Therefore the worst  $p$  occurs at either endpoint.

**Theorem AQ4 (necessary and sufficient criterion in this family).** Define

$$p_{N,m}(\beta) = \sum_{j=0}^{m-1} \binom{N}{j} (\cos^2 \beta)^j (\sin^2 \beta)^{N-j}.$$

For  $0 < \epsilon < 1$ , the ideal uniform complete-channel error is at most  $\epsilon$  exactly when

$$2p_{N,m}(\beta) - p_{N,m}(\beta)^2 \leq \epsilon^2,$$

equivalently  $p$  is at most  $1 - \sqrt{1 - \epsilon^2}$ . The clock-traced requirement instead uses  $p \leq \epsilon$ .

*Proof.* The binomial monotonicity above locates the endpoint maximum. The exact channel formulas in AQ2-AQ3 are monotone functions of  $p$  on  $[0,1]$ . Solving the quadratic inequality gives the equivalent criterion. Both endpoints are included, so necessity is attained within the declared window.  $\square$

This finite sum makes the integer resource frontier exactly computable. It is a necessary-and-sufficient clock-size condition for this family, not a prediction of device fault tolerance.

### 5.1 Matching asymptotic size scaling for the fixed-width convention

Set  $\beta = \pi/4$ . Then

$$p_{N,m} = \frac{1}{2^N} \sum_{j=0}^{m-1} \binom{N}{j}.$$

**Corollary AQ5 (scoped clock-size scaling).** For  $m \geq 1$  and  $0 < \epsilon < 1/2$ , the least  $N$  in this fixed- $\beta$  family satisfying the ideal complete-error requirement has order

$$N_{\min} = \Theta(m + \log(1/\epsilon)).$$

*Proof of necessity.*  $N \geq m$  is required by the gate word. The tail includes its address-0 term, so  $p \geq 2$  to the power  $-N$ . Since  $2p - p^2 \geq p$ , error  $\leq \epsilon$  implies  $2$  to the power  $-N \leq \epsilon^2$ . Thus  $N \geq 2 \log_2(1/\epsilon)$ . The maximum of these two lower bounds is at least half their sum.

*Proof of sufficiency.* If  $K$  has  $\text{binomial}(N, 1/2)$  distribution, Markov's inequality applied to  $3$  to the power  $-K$  gives

$$p = \Pr(K \leq m-1) \leq 3^{m-1} \left(\frac{2}{3}\right)^N.$$

For  $N \geq 4(m-1)$ , this is at most  $\exp(-\eta N)$ , with

$$\eta = \frac{3}{4} \ln 3 - \ln 2 > 0.$$

Because  $\sqrt{2p - p^2} \leq \sqrt{2p}$ , it is sufficient to choose

$$N \geq \max \left\{ m, 4(m-1), \left\lceil \frac{\ln(2/\epsilon^2)}{\eta} \right\rceil \right\}.$$

The constants are independent of  $m$  and  $\epsilon$ . Necessity and sufficiency give the stated order within this family.  $\square$

Clock Hilbert-space dimension is  $N + 1$ . Unary single-excitation physical encoding uses  $N + 1$  sites. Binary encoding uses  $\lceil \log_2(N + 1) \rceil$  address qubits but has different coupling-synthesis requirements. The scaling above concerns  $N$ , not a universal lower bound on physical clock qubits. It applies to the stated broad normalized window; a single exact midpoint or a narrower window is a different task.

## 5.2 Rate, latency, interaction norm and locality

On the legal address sector, the clock has spectrum  $\kappa(N - 2j)$ , so its spectral norm is  $g = \kappa N$ . At fixed  $g$  and  $\beta = \pi/4$ ,

$$T = \mathcal{H} = \frac{\pi N}{2g}.$$

Increasing  $N$  improves the ideal tail while increasing launch-to-midpoint latency and readout duration at fixed  $g$ . The result is not a minimum-time theorem across architectures. If latency or duration is prescribed independently, use the general  $\beta$  criterion rather than substituting the fixed-width corollary unchanged.

For a unary single-excitation encoding, each clock edge can be written as a hopping term on two neighboring clock sites multiplied by its body gate, plus the Hermitian conjugate. A one-body-qubit primitive produces a three-local interaction; a two-body-qubit primitive produces a four-local interaction. Each edge term has norm  $\kappa \sqrt{(j+1)(N-j)}$ . This supplies a bounded-body-support implementation formula on the legal sector. It does not establish two-local synthesis, geometric locality of all body connections, bounded scalable interaction degree or hardware feasibility. The norm  $g$  is not asserted to bound the whole many-excitation physical Hilbert space. Leakage out of the legal clock sector requires another admitted error model.

Initial clock address 0 has zero mean energy under this generator and variance  $\kappa^2 N$ , independently of the body input. This follows from off-diagonal address support and the first edge's squared norm. At fixed  $g$  the variance is  $g^2/N$ . The clock is still nonstationary; zero mean in this energy convention does not make preparation free. Total energy under the supplied constant Hamiltonian is conserved, but no dissipative cost or resource-preparation mechanism is derived.

## 5.3 A latency bound for the broader nearest-neighbor history class

Consider any address-0 history chain with arbitrary diagonal clock potentials and adjacent hopping amplitudes of magnitude at most  $J_{\max}$ . Prefix dressing, the final exhaustion flag and the chosen complete target are as above. Let  $Q_C = \sum_j j |j\rangle\langle j|$  be the clock-position operator.

**Proposition AQ7 (nearest-neighbor latency obstruction).** If the exact complete target error at time  $t$  is at most  $\epsilon < 1$ , then

$$t \geq \frac{m\sqrt{1-\epsilon^2}}{2J_{\max}}.$$

*Proof.* Diagonal potentials commute with  $Q_C$ . The commutator with nearest-neighbor hopping has row and column absolute sums bounded by  $2J_{\max}$ , and hence operator norm at most  $2J_{\max}$ . Prefix dressing commutes with  $Q_C$  and preserves this bound. Therefore the initial position zero implies  $\langle t | Q_C | t \rangle \leq 2J_{\max} t$ . Meanwhile the completed sector  $j \geq m$  has weight  $1 - p$ , so  $\langle t | Q_C | t \rangle \geq m(1 - p)$ . AQ2 remains valid for any clock amplitudes under the same flag and prefix contract. Error  $\leq \epsilon$  thus implies  $1 - p \geq \sqrt{1 - \epsilon^2}$ , which gives the result.  $\square$

This is a coarse propagation bound for nearest-neighbor history chains, not a new universal quantum-computing speed limit. It excludes direct shortcut interactions outside the assumed address graph. At fixed  $\epsilon$  the engineered construction with  $N$  of order  $m$  has the same linear order in  $m/J_{\max}$ , but a finite-width clock-size lower bound spanning all such chains has not been proved. The numerical comparison below does not close that gap.

## 6. Complete bank admission and robustness

Suppose the implemented static generator on the same legal joint domain is  $H_{\text{hist}} + \Delta$ , with  $\text{norm}(\Delta) \leq d$ . Let the actual supplied bank and input have complete trace distance at most  $\epsilon_B$  from the specified input/reference state times the ideal prepared bank. All outputs, donors, records and references stay in the comparison; partial trace is used only for an explicitly reduced diagnostic.

**Theorem AQ6 (unconditional complete budget).** At every  $t \geq 0$ ,

$$D(\rho_t^{\text{act}}, \rho_t^{\text{target}}) \leq \min\{1, \epsilon_B + td + \sqrt{2p(t) - p(t)^2}\}.$$

On  $I_{\text{beta}}$ , a uniform upper bound substitutes  $t = (\pi/2 + \text{beta})/\kappa$  and the endpoint tail. With  $\text{beta} = \pi/4$  and  $g = \kappa N$ ,

$$\epsilon_{\text{complete}} \leq \min\left\{1, \epsilon_B + \frac{3\pi Nd}{4g} + \sqrt{2p_{N,m} - p_{N,m}^2}\right\}.$$

*Proof.* Unitary contractivity propagates the initial complete supply discrepancy without increasing it. Duhamel bounds the actual/comparison evolution distance by  $td$  on all spectator extensions. AQ2 bounds comparison versus the stated target. The triangle inequality and the trace-distance ceiling give the result. The endpoint tail and maximal elapsed time give the uniform version.  $\square$

For a fixed ideal bank this is also a channel upper bound with  $\epsilon_B = 0$ . An  $\epsilon_B$  channel term requires a uniform completely bounded preparation-channel guarantee; a single supplied-state distance does not automatically become a diamond norm. Extra integrated perturbations can be admitted on the same domain. A probability diagnostic already covered by complete state distance is not an additional independent error charge.

This robustness bound is sufficient. It is not accompanied by a matching necessary mismatch-cost theorem. In particular, failure of this upper bound to certify a budget is not proof that the actual device must fail.

For a fresh ideal exhaustion bit, the ideal dynamical comparison has NO\_COMPLETE probability  $p$ . The false-tag event is  $E = 1$  while clock address  $j < m$ ; it has probability zero in that comparison. Projecting purifications and using the same Duhamel amplitude bound gives the actual diagnostics

$$\begin{aligned} \Pr(\text{NO\_COMPLETE}) &\leq \min\{1, \epsilon_B + (\sqrt{p} + td)^2\}, \\ \Pr(\text{FALSE\_TAG}) &\leq \min\{1, \epsilon_B + (td)^2\}. \end{aligned}$$

Initial supply discrepancy changes probabilities linearly by at most  $\epsilon_B$ . These are consequences of the existing joint comparison and are not additional independent state-error debits. A numerical flag-flipping perturbation supplies a nonzero false-tag fixture. The ideal dynamical comparison here is the history state, whereas the complete target in AQ2 is the completed body times its comparison clock; the two comparisons should not be conflated.

## 7. Exact finite example and numerical verification

For  $m = 13$  and target  $\epsilon = 1/1024$  under the fixed-width convention, exact rational enumeration finds  $N_{\text{min}} = 63$ . Thus the clock has 64 address states. At either window endpoint,



$$p = \frac{3440189695001}{9223372036854775808}, \quad \epsilon_{\text{joint}} = \sqrt{2p - p^2} < \frac{1}{1024}.$$

$N = 62$  fails the same complete-channel criterion. The approximate views are  $p = 3.72986$  times  $10$  to the power  $-7$  and complete error  $8.63697$  times  $10$  to the power  $-4$ . They are model diagnostics, not measured error rates. The exact resource decisions use rational  $p$  and the squared-error inequality, avoiding a rounded square-root threshold.

| Target complete error | Least $N$ | Clock addresses | Exact-test status   |
|-----------------------|-----------|-----------------|---------------------|
| 1/2                   | 32        | 33              | Pass; $N - 1$ fails |
| 1/4                   | 37        | 38              | Pass; $N - 1$ fails |
| 1/8                   | 41        | 42              | Pass; $N - 1$ fails |
| 1/16                  | 44        | 45              | Pass; $N - 1$ fails |
| 1/32                  | 48        | 49              | Pass; $N - 1$ fails |
| 1/64                  | 51        | 52              | Pass; $N - 1$ fails |
| 1/128                 | 54        | 55              | Pass; $N - 1$ fails |
| 1/1024                | 63        | 64              | Pass; $N - 1$ fails |

Take  $g = 1$ ,  $\kappa = 1/63$ ,  $d = 1/(2 \text{ to the power } 21)$  and  $\epsilon_B = 1/(2 \text{ to the power } 16)$ . The resulting uniform sufficient complete error is below  $1/1024$  over

$$\frac{63\pi}{4} \leq t \leq \frac{189\pi}{4}.$$

Its duration is  $63\pi/2$  in model units. The initial bank term is a supplied certification allowance. The numerical dynamics below use the ideal prepared bank and a specified local work perturbation  $d_{Z_A}$ ; they do not physically measure or synthesize a bank with that allowance. Six binary clock qubits encode the 64 addresses; the body has nine qubits, and the fixture adds one spectator qubit. A unary clock realization instead has 64 clock sites. Their implementation costs are not equated.

The replay uses a Bell-correlated input and two additional seeded pure input/reference states. It constructs body gates, checks each unitary, checks exhaustion-sector orthogonality, compares the exact complete and reduced errors on 72 analytic time/input/clock fixtures, and independently propagates sparse Hamiltonians at  $N = 13, 24$  and  $63$ . The tested clock indices include unpadded, insufficiently padded and sufficiently padded cases. Diagonalizing the small clock verifies the admitted norm. A local static perturbation checks the full error budget.

The final active module has its declared fresh state. The first outgoing donor junk qubit retains one bit of mutual information with the input spectator, and the second outgoing donor retains about 1.20175 bits with output 2. The latter output/donor pair is quantum-correlated rather than a fresh independent resource. A deliberately correlated donor gives two bits of active/spectator mutual information after exchange, rejecting the independent-reset inference. Projective return undoes  $E$ , rejecting permanent exhaustion reporting.

The second output/donor pair has a partial-transpose eigenvalue  $-\sqrt{2}/4$ , independently confirming entanglement; mutual information alone would not establish it.

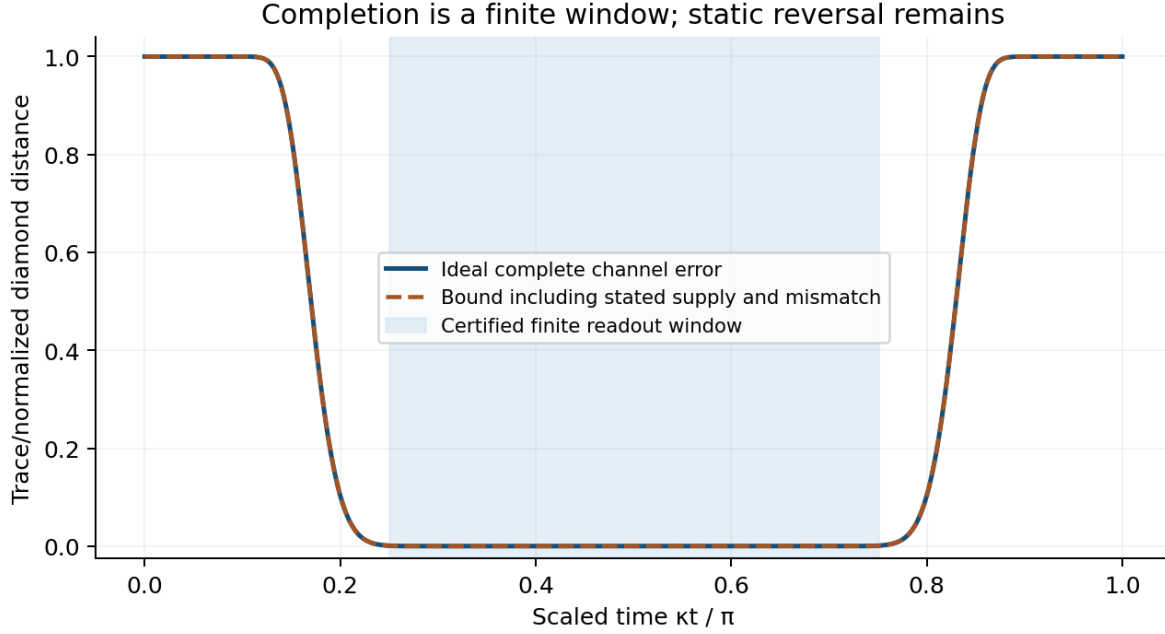


Figure 2: Finite readout and subsequent return at  $N = 63$  and  $g = 1$ . The robustness curve uses the stated mathematical supply and mismatch allowances.

### 7.1 A matched-norm clock baseline

A uniform path clock is compared with the engineered spin clock at  $N = 63$  and admitted spectral norm  $g = 1$ . Both start at address 0, use the same thirteen-step body word and prepared bank, and are scored against their own complete-word comparison clock over the same requested observation interval. The uniform edge strength is  $1/(2 \cos(\pi/65))$ , giving its admitted norm exactly one.

The comparison reports 1,001 time samples per clock. The uniform path fails the  $1/1024$  complete-error requirement at sampled times; the spin-clock window has the exact binomial certificate above. The uniform grid is not an optimized search over all clocks or all time windows. Clock preparation variance and edge norms differ and are explicitly reported: equal spectral norm does not make every resource equal. Hence the benchmark is a specified baseline, not a resource-optimality or quantum-advantage theorem.

The tail derivative satisfies  $|p'(t)| \leq |a_{(m-1)}|$ , where  $a_{(m-1)}$  is the one clock edge crossing the incomplete/completed cut. Only that edge contributes to the cut-projector commutator. Half the grid spacing times this edge norm is therefore an analytic sampling allowance for the maximum tail. The reported sampling bound uses numerical grid values; it is not an interval-arithmetic certificate for their numerical error.

The executed receipt records **32 exact rational controls and 2313 numerical checks**, plus 7 qualitative numerical negative/threshold controls, with tolerance  $2e-10$  and maximum residual  $2.50028e-13$ . Exact controls concern rational frontier decisions and the Markov bound; numerical checks include multiple diagnostics on shared fixtures and repeated grid points. These counts are not independent experiments.

These are constructor fixtures. The general channel equalities and scaling rely on the written proofs, not enumerating finite states. A finite fixture count does not establish independent review or device

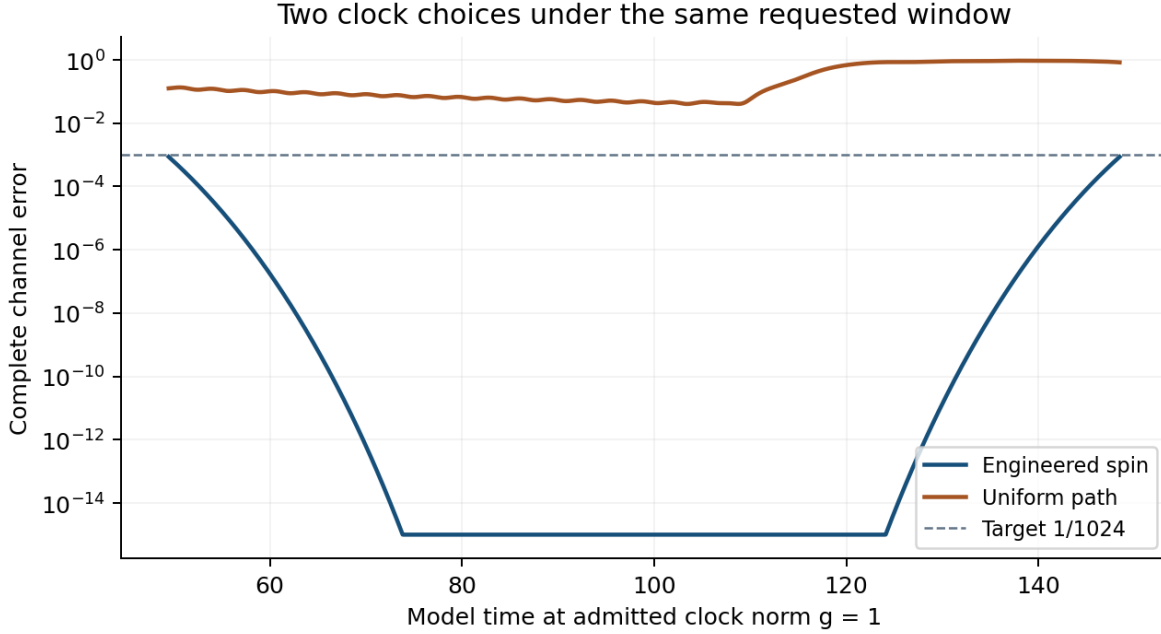


Figure 3: Specified uniform-path and spin-clock baseline under the same norm, word, input address and observation interval. The spin curve uses its analytic binomial law; errors below  $1e-15$  are drawn at the plotting floor. Other resource differences are retained in the comparison receipt.

feasibility.

## 8. Prior-art comparison and the next theorem frontier

Christandl et al. [1] supply engineered perfect transfer. Caha, Landau and Nagaj [2] study history clocks and several idling constructions, including ground-state amplification and dynamical clock variants. Woods, Silva and Oppenheim [3] study finite-clock control and backreaction in a different construction. These works establish important ingredients; a longer idle suffix is not by itself a new autonomous-computation principle.

Meier et al. [4] formulate an autonomous quantum processing unit using an instruction register, clock and memory in an open thermal framework. Our compiled closed finite episode has different domains and does not reproduce their thermodynamic analysis. Woods [5] studies coherent-control runtime advantages under power-resource assumptions; this addendum establishes no such advantage. A direct theorem-level comparison with these methods remains pending.

The present advancement is the explicit finite episode, its exact complete-versus-reduced channel relation, and an exact integer frontier plus matching size order within a stated family. The additional path-latency obstruction is coarse and the uniform-clock comparison is a baseline, not broader optimality. The global supervisor remains an unrenowned resource. The original fuel/reference preparation gates, full nondegenerate energy constraints, continuous apparatus realization, geometric implementation and independent proof review remain open.

The next strongest research target is a comparison across admissible clock families at common gate set, local interaction limits, launch latency, readout duration, retained-domain target and supply quality. Seek either a strict resource improvement that survives complete accounting or a lower bound spanning

that broader class. The family-specific size law derived here is a baseline for that effort, not a substitute for it. Completing an autonomous refresh of the supervising clock while accounting for exported correlations is a separate target; swapping the work module does not solve it.

## Reproduction and provenance

Run the accompanying replay with `OPENBLAS_NUM_THREADS=1` and Python 3 with NumPy and SciPy. The exact frontier, time sweep, perturbation fixtures, compiled word and replay receipt are written to `results/`. General input `Y` can have arbitrary dimension in the proofs; the numerical fixture uses one reference qubit. The parent manuscript and release are identified by hashes in `INPUT_ANCHORS.json`. They are not modified by this addendum. Retain constructor checking, independent proof review, novelty and physical validation as separate statuses.

## References

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